GeoGebra – 8th Standard

ALGEBRA

Solving a Simple Linear Equation

This activity demonstrates how to solve a linear equation interactively in GeoGebra. The values of a, b, and can be changed, and the intermediate calculation and answer update automatically.

Question:

Find the value of x in the linear equation 2x + 3 = 11.

Step 1: Open GeoGebra

1. Open GeoGebra.

2. Choose Calculator Suite.

3. Open the Algebra view and locate the Input Bar.

Step 2: Create three numbers

In the Input Bar, enter these commands one by one:

```
a = 2
b = 3
c = 11
```

These values represent the equation ax + b = c. For this example, the equation is 2x + 3 = 11.

Step 3: Calculate the intermediate value and answer

Enter the following commands:

```
d = c - b
ans = d / a
```

Here, d = c − b and x = d/a, provided a ≠ 0. For the example, d = 8 and the answer is x = 4.

Step 4: To calculate the value of x we have four steps

Step 4.1: Add or subtract b from both sides, depending on its sign

Enter this command in the Input Bar:

```
S1 = Text(If(b > 0, "Step 1: Subtract " + b + " from both sides", If(b < 0, "Step 1:
Add " + abs(b) + " to both sides", "Step 1: No constant to remove")))
```

Step 4.2: Simplify to ax = c - bS

Enter:

```
S2 = Text( a+ "x = " + d)
```

For the example, the simplified equation is 2x = 8.

Step 4.3: Divide both sides by a

Enter:

```
S3 = Text("Step 2: Divide both sides by " + a)
```

This explains how to isolate x.

Step 4.4: Display x = (c – b) / a provided a ≠ 0

Enter:

```
S4 = Text("Answer: x = " + ans)
```

For the example, the answer is x = 4.

Step 5: Create input boxes

Definition: An Input Box is an interactive tool in GeoGebra that allows user to enter, edit or modify the value of a linked mathematical object directly on the screen.

1. Open Tools and select Input Box.
2. Click in the Graphics view to place an input box.
3. In the dialog box, link the input box to the existing object a.
4. Repeat the process for b and c.

Changing a, b, or c should automatically update d and ans. The exact dialog labels may vary slightly by GeoGebra version.

Step 6: Try different equations

Change the input values to try these examples:

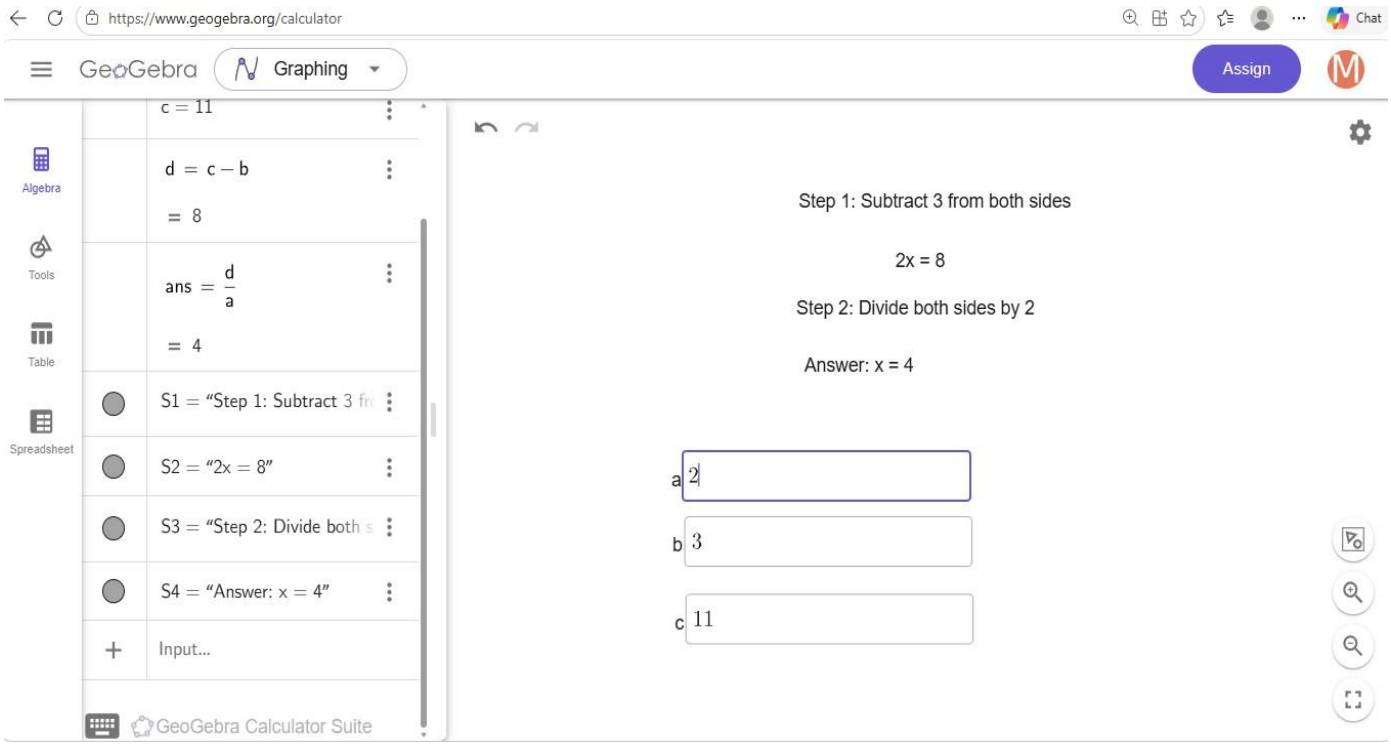
- 3x + 6 = 15, here a = 3, b = 6, c = 15
- 5x − 7 = 8
- 4x + 2 = 18

Important notes

• Ensure that a ≠ 0, because division by zero is undefined.

• The Text commands create visible text objects; they do not create clickable buttons.
• To reveal Step 1, Step 2, Step 3, and the answer one at a time, add buttons and visibility controls as a separate stage.

Output:



GeoGebra – Product of Monomials

The general form:

$$(c_1 m^{p1} n^{q1})(c_2 m^{p2} n^{q2}) \ \ldots \ (c_r m^{pr} n^{qr})$$

Question:

Find the Product of the terms $-2mn,\ 2m^2, -3mn$

Step 1: Enter the coefficients

Open GeoGebra and choose Calculator Suite. In the Input Bar, enter the three numerical coefficients:

```
c1 = -2
c2 = 2
c3 = -3
```

Step 2: Enter the exponents of m

```
p1 = 1
p2 = 2
p3 = 1
```

Step 3: Enter the exponents of n

```
q1 = 1
q2 = 0
q3 = 1
```

Step 4: Calculate the product of the coefficients

```
K = c1 * c2 * c3
```

Step 5: Calculate the total exponent of m

```
P = p1 + p2 + p3
```

Step 6: Calculate the exponent of n

```
R = q1 + q2 + q3
```

Step 7: Display the coefficient calculation

```
S1 = Text("Step 1: Coefficient product = " + K)
```

Step 8: Display the exponent calculation for m

```
S2 = Text("Step 2: Exponent of m = " + p1 + " + " + p2 + " + " + p3 + " = " + P)
```

Step 9: Display the exponent calculation for n

```
S3 = Text("Step 3: Exponent of n = " + q1 + " + " + q2 + " + " + q3 + " = " + R)
```

Step 10: Display the final answer

```
S4 = Text("Final answer: " + K + "m^" + P + "n^" + R)
```

Make the construction reusable with input boxes

Create input boxes so the coefficients and exponents can be changed on the screen.

1. Open Tools and select Input Box.
2. Place an input box in the Graphics view.
3. Set Caption = c1 and Link Object = c1.
4. Repeat for c2, c3, p1, p2, p3, q1, and q2, q3.

When linked values change, the calculations and displayed text should update automatically.

Example values from the notes

Coefficients: c1 = −2, c2 = 2, c3 = −3.

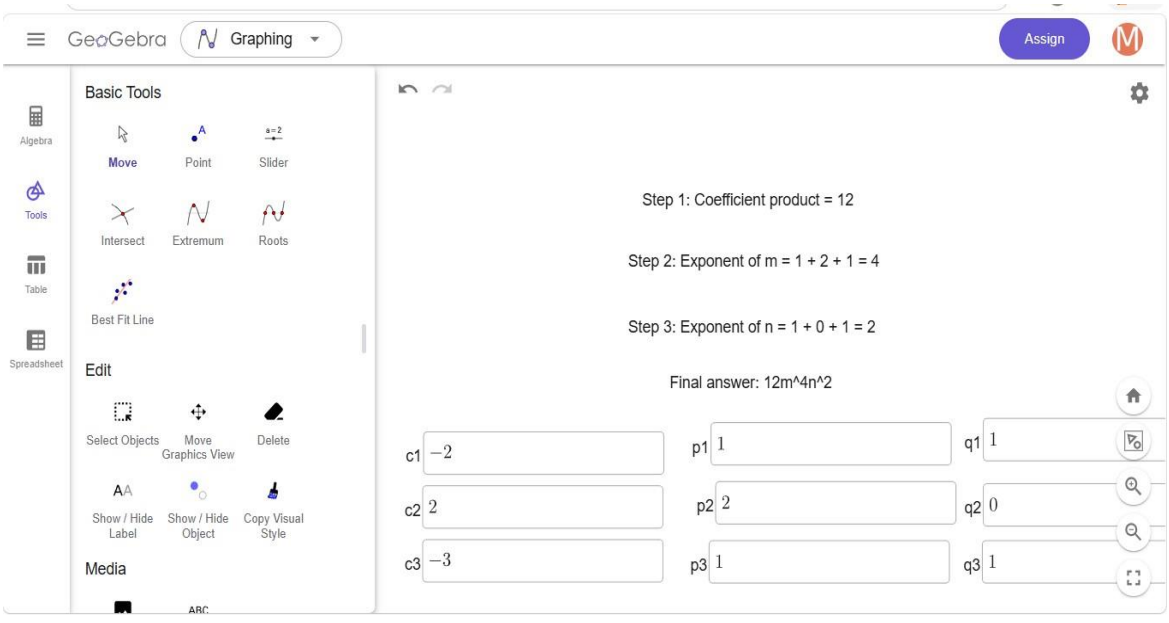
Exponents of m: p1 = 1, p2 = 2, p3 = 1.

Exponents of n: q1 = 1, q2 = 0, q3 = 1.

The coefficient product is K = (−2)(2)(−3) = 12.

Using the listed exponents, the product is $12m^4n^2$.

Output:



Expansion of $(x + a)^3$

Question:

Expand $(x + 4)^3$ using the cubic identity $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.

Open GeoGebra and choose Calculator Suite.

Step 1: Create a Slider

```
a = 4
```

Minimum: 1

Maximum: 20

Increment: 1

Step 2: Display the expression

```
Question = Text("Explore (x + a)^3, where a = " + a)
```

Step 3: Display the identity

```
Identity = Text("(x + a)^3 = x^3 + 3ax^2 + 3a^2x + a^3")
```

Step 4: Show the expansion terms separately

```
T1 = Text("1. First term: x^3")
```

```
T2 = Text("2. Second term: 3 a x^2 =" + (3*a) + "x^2")
```

```
T3 = Text("3. Third term: 3 a^2x= " + (3*a^2) + "x")
```

```
T4 = Text("4. Last term: a^3= " + a^3)
```

Step 5: Prepare the final answer

```
FinalAnswer = Text("Final Answer: x^3 + " + (3*a) + "x^2 + " + (3*a^2) + "x + "
+ a^3)
```

Step 6: Create a button to reveal the answer

1. Find the Button tool in the Tools menu.
2. Click in the Graphics View where you want the button

Caption : Answer

GeoGebra Script: `SetVisibleInView(FinalAnswer, 1, true)`

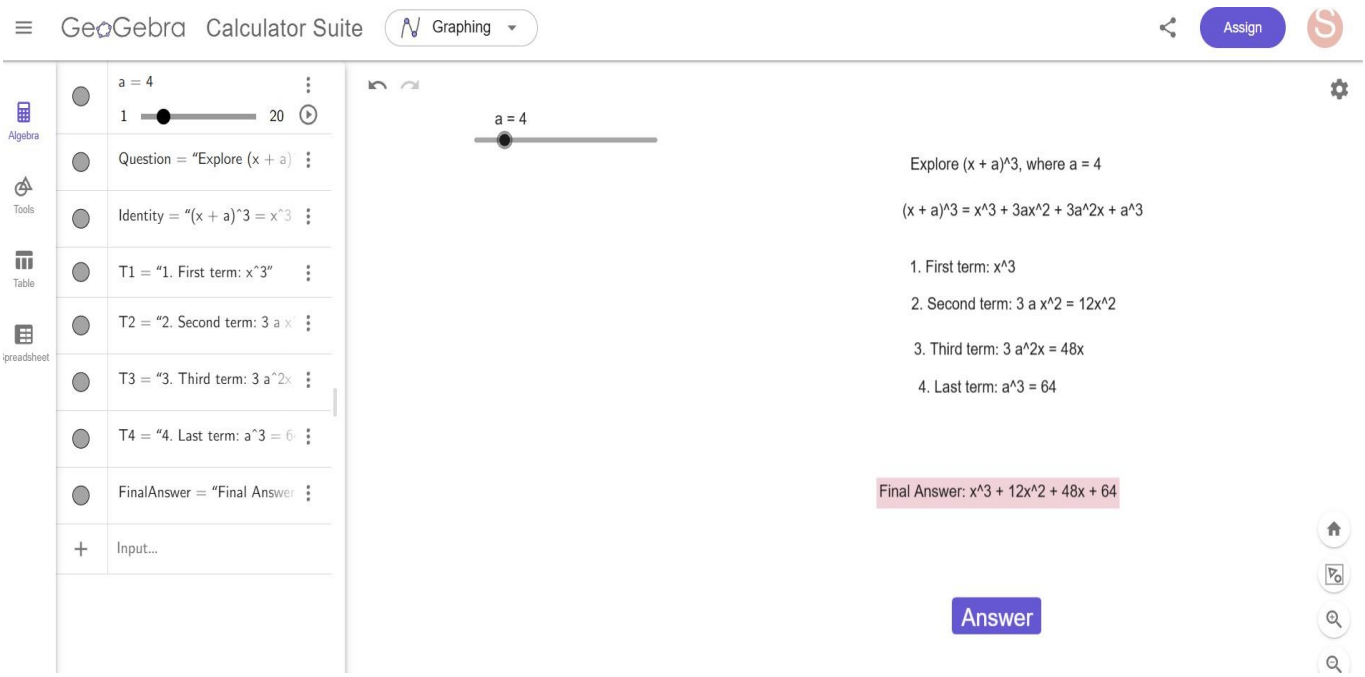
Note about the button

If the final answer should remain hidden until the button is clicked, first hide FinalAnswer in the Graphics View or set its visibility to false. Then use the button script to make it visible.

How to use the demonstration

- Change the slider a from 1 to 20.
- Observe how the coefficients 3a, $3a^2$ and the constant term a^3 change.
- Click the Answer button to reveal the final expansion.
- Compare the displayed result with $(x + a)^3 = x^3 + 3ax^2 + 3a^2x + a^3$.

Output:



Graphing Linear Equations

Open GeoGebra and choose Calculator Suite.

Question:

Draw the graph of the linear equation y = -x + 5.

Step 1: Display the question number

```
Text("Question number: " + q)
```

Create a slider named q with integer values from 1 to 9.

Step 2: Display the corresponding question

```
Text("Question: " + If(q == 1, "y = 2x + 5",
If(q == 2, "y = x + 3",
If(q == 3, "y = 3x + 2",
If(q == 4, "y = 2x - 4",
If(q == 5, "y = -x + 5",
If(q == 6, "y = 4x + 1",
If(q == 7, "y = -2x + 3",
If(q == 8, "y = 5x - 2",
If(q == 9, "y = 3x - 6", "y = x - 2"))))))))))
```

Step 3: Define f(x)

```
f(x) = If(q == 1, 2x + 5,
 If(q == 2, x + 3,
 If(q == 3, 3x + 2,
 If(q == 4, 2x - 4,
 If(q == 5, -x + 5,
 If(q == 6, 4x + 1,
 If(q == 7, -2x + 3,
 If(q == 8, 5x - 2,

 If(q == 9, 3x - 6, - x - 2)))))))))
```

Step 4: Create a table to display x and y values

```
xValues = {-2, -1, 0, 1, 2}
```

```
yValues = {f(-2), f(-1), f(0), f(1), f(2)}
```

The y-values are calculated automatically from the selected function.

Step 5: Display the coordinate pairs

```
P = {(-2, f(-2)), (-1, f(-1)), (0, f(0)), (1, f(1)), (2, f(2))}
```

```
Sequence(Text(Element(P, i),Element(P, i)), i, 1, 5)
```

How to use the demonstration

- Move the slider q from 1 to 9 to select a different question.
- Read the displayed equation and observe how the function changes.
- Check the x-values −2, −1, 0, 1, 2 and their corresponding y-values.
- Observe the coordinate pairs generated for the selected function.

Output



Product of a Monomial by a Polynomial

Question:

$(3x^2y)(2x + 5y^2)$

General Forms:

Monomial: ax^ny^m

Polynomial: $px^r + by^s$

Open GeoGebra and choose Calculator Suite.

Step 1: Enter the values

In the Input Bar, enter the following values:

```
a = 3
p = 2
b = 5
n = 2
m = 1
r =1
s = 2
```

Step 2: Create input boxes

Create Input Boxes for a, p, b, n, m, r and s. These allow you to change the coefficients and exponents.

Step 3: Display the question

Use a Text command to display the expression to be multiplied:

```
Question = Text("Multiply: " + a + "x^" + n + "y^" + m + " (" + p + "x^" + r + " + " + b + "y^" + s + ")")
```

Step 4: Display the distributive property

```
Text("Multiply the monomial by each term inside the bracket.")
Text( a + "x^" + n + "y^" + m + " (" + p + "x^"+ r + "+" + b + "y^" + s + ")")
```

Step 5: Display the law of exponents

```
Law = Text("Law of Exponents: x^m * x^n = x^(m+n)")
```

Step 6: Display the question

```
Question=Text("Predict the missing exponent: x^"+m+" * x^"+n+" = x^?")
```

Step 7: Create an answer input

```
ans = 0
```

Create an Input Box and link the object to ans.

Step 8: Create a Boolean variable

Create a Boolean variable and set its initial value:

```
check = false
```

Step 9: Create feedback text

```
Feedback = If(check, "Correct!! Well done!!", "Try again!")
```

Step 10: Update the button script

In the button's On Click script, enter:

```
SetValue(check, ans == m + n)
```

Step 11: Display the final result

```
FinalAnswer = Text("Answer: "+(a p)+"x^"+(n+r)+"y^"+m+" + "+(a b)+"x^"+n+"y^"+(m+s))
```

Output

